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EXPERIMENT 6: Jira Agile Project Management

AIM

To create and manage a Scrum-based software development project using Jira by creating epics, user stories, tasks, sprints, issue dependencies, and an active project board.

THEORY

Jira is an Agile project management tool used to plan, track, and manage software development activities. In the Scrum methodology, a project can be organized into different levels such as **Roadmap, Epics, User Stories, and Tasks/Sub-tasks**.

An **Epic** represents a large project initiative, while **User Stories** describe functional requirements that provide customer value. **Tasks** represent specific activities required to complete the work.

Jira also provides a **Backlog** for managing issues and a **Sprint** for selecting a set of tasks to be completed within a specific time period. Issues can be linked to show dependencies between tasks. The active sprint is managed using a board with columns such as **To Do, In Progress, Review, and Done**.

PROCEDURE

1. Create Jira Account

1. Open Jira and select **Get Jira for free**.
2. Register using an email address and verify the account.
3. Enter the required account details.
4. Select **Software Development** as the work type.
5. Select **Scrum** as the methodology.

2. Create Project

1. Create a new Scrum project.
2. Enter the project name as **Parking Management System**
3. Open the created project.

3. Add Team Members

1. Open the **Team/People** section.
2. Enter the email addresses of collaborators.
3. Set the required permissions.
4. Click **Add people**.

4. Create Epics

Create the following two epics:

- **Improve Customer Service**
- **Increase Product Page Conversions**

5. Create Issues

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Issue Type	Summary	Parent Epic
Story	Create user login and registration interface	User Authentication & Interface
Task	Set up parking database schema	Core Parking Engine & Infrastructure
Task	Create parking fee calculation logic	Core Parking Engine & Infrastructure
Story	Make UI	User Authentication & Interface
Story	Add backend	Core Parking Engine & Infrastructure

6. Create Issue Dependencies

- The **Make UI** and **Add backend** tasks are dependent on the completion of the **Set up parking database schema** task.
- The **Connect APIs** task is dependent on **Make UI** and **Add backend**.
- Use the **Link issue** option and select **is blocked by** / **blocks** to establish these dependencies.

7. Configure Sprint

- **Open the Backlog.**
- **Select SCRUM Sprint 0 (or Sprint 1).**
- **Edit the sprint details.**
- **Set the sprint dates from 24 Sep to 8 Oct.**
- **Set the sprint goal as:**

"Develop core infrastructure, user authentication, and parking calculation logic."

- **Move the following issues into the sprint:**
  - **Create user login and registration interface**
  - **Set up parking database schema**
  - **Create parking fee calculation logic**
  - **Make UI**
  - **Add backend**
  - **Connect APIs**

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- **Containerize using Docker**
- **Add Monitoring using prometheus and grafana**

8. Start Sprint

1. Click **Start Sprint**.
2. Confirm the sprint.
3. The sprint will open as an **Active Board**.

9. Add Review Column

1. Open the active sprint board.
2. Add a new column named **Review**.
3. Place it between **In Progress** and **Done**. The final workflow becomes:

To Do → In Progress → Review → Done

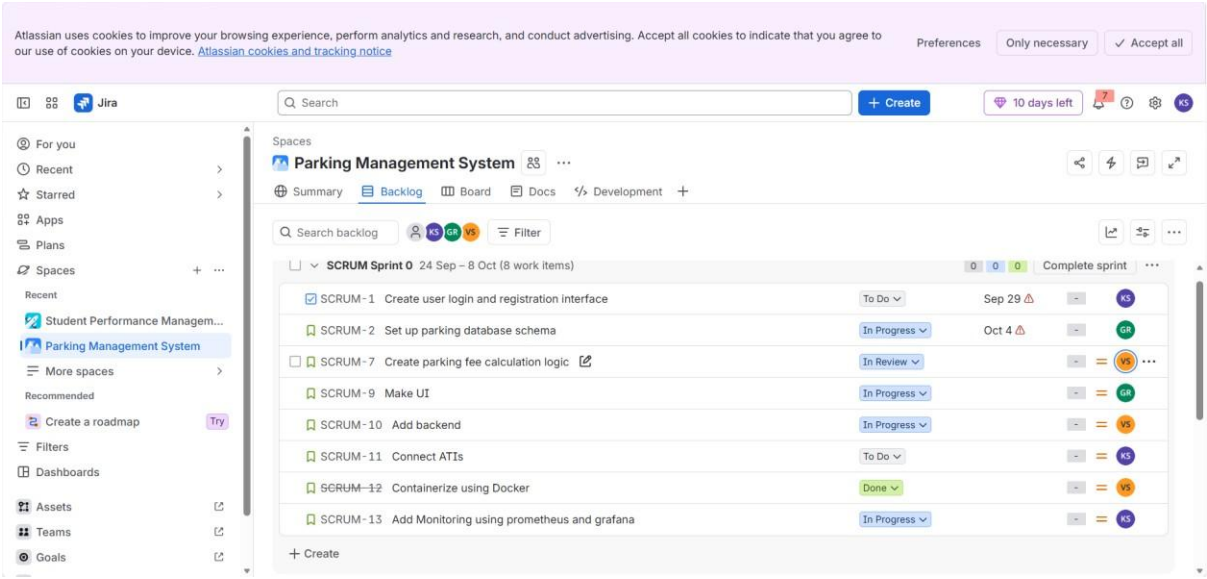
10. View Project Information

The project can also be viewed using:

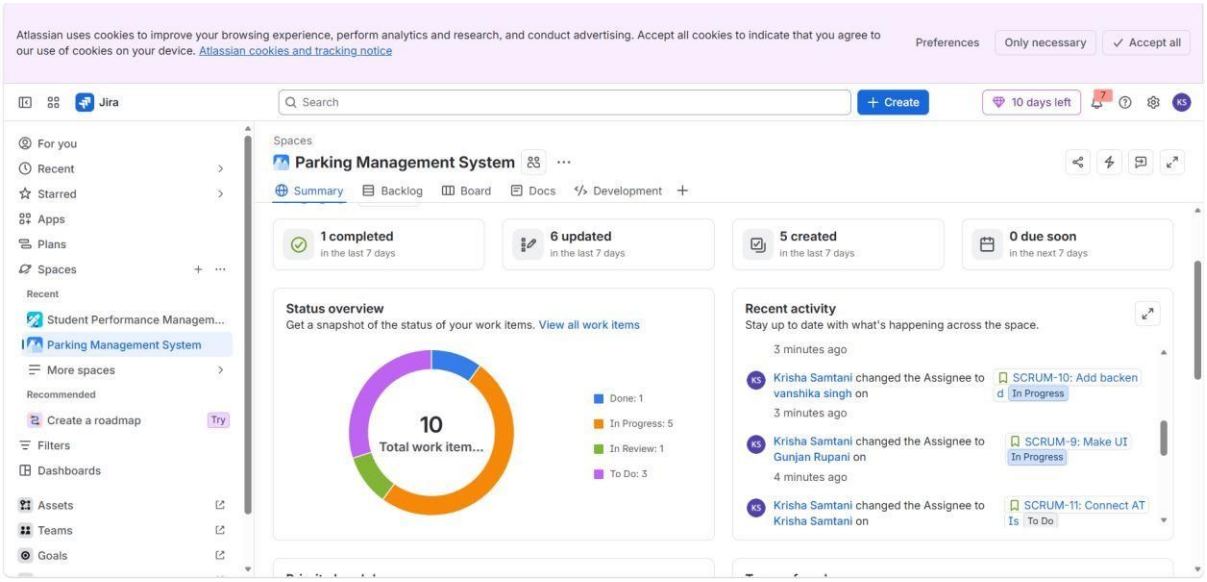
- **Timeline View** – displays epics, tasks, dates, and dependencies.
- **List View** – provides a spreadsheet-style view of issues.
- **Dashboards** – provide project tracking and reporting.

RESULT

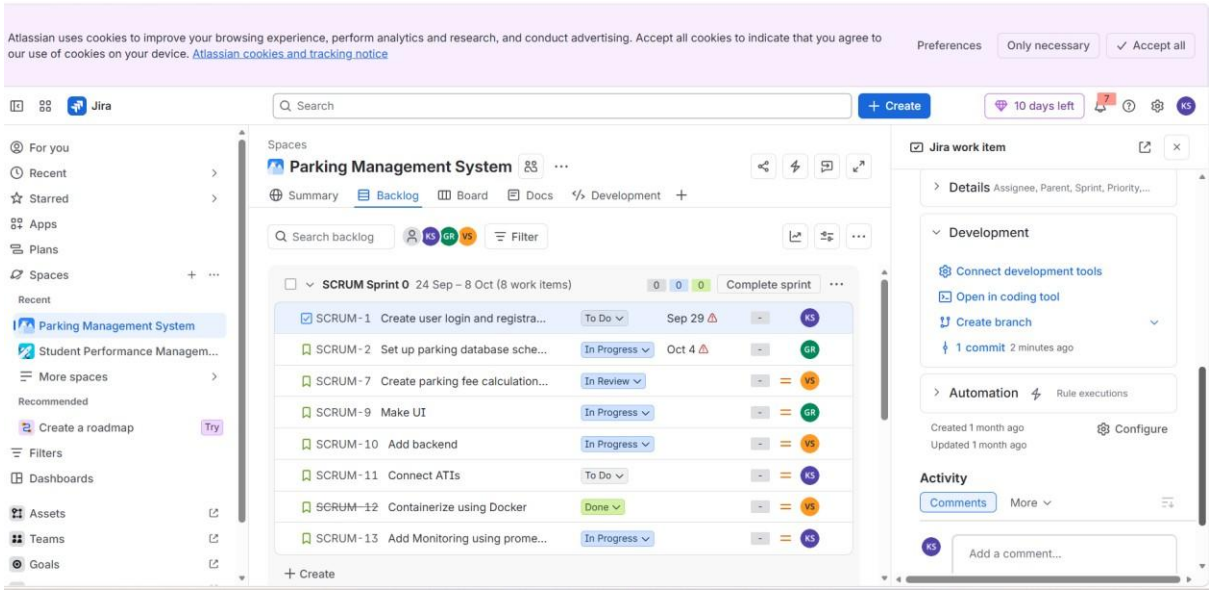
The **Parking Management System** Scrum project was successfully created in Jira. Epics, user stories, backend and frontend tasks, issue dependencies, and **SCRUM Sprint 0** were configured successfully. The sprint was started, team members were assigned, and a custom **In Review** column was added to the active board.



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CONNECTION TO GITHUB



CONCLUSION

Thus, Jira was successfully used for **Agile Scrum project management**, including project planning, backlog management, sprint planning, issue tracking, dependency management, and progress monitoring.